

Article

Simulation and Dynamic Assessment of Urban Water Metabolism Using an ABM–SD-Coupled Model: A Case Study of Beijing

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Abstract

Water-scarce megacities undergoing rapid urbanization and climate change often face imbalances in urban water metabolism, characterized by the coexistence of socioeconomic growth and increasing pressure on water environmental carrying capacity. To address the limitations of existing studies in capturing the dynamic evolution of urban water systems, this study develops an urban water metabolism simulation model based on an agent-based modeling and system dynamics (ABM–SD)-coupled framework. The model represents the system evolution mechanism of “micro-level decision making–macro-level feedback” by incorporating the behavioral interactions of multiple agents, including the government, residents, and enterprises. On the basis of model simulation, the coupling coordination degree (D) and the Nemerow index (PI) are introduced to construct a two-dimensional D–PI state space, thereby enabling the dynamic assessment of the coordination and sustainable development status of the urban water metabolism system. Taking Beijing as a case study, the simulation results show that the coordination degree between socioeconomic development and the water environment continues to improve, while pressure on the water environmental carrying capacity continues to accumulate. Overall, the system exhibits a non-synchronous evolutionary pattern characterized by improved coordination alongside increasing carrying-capacity risks.

Keywords: urban water metabolism; ABM–SD-coupled model; system dynamics; agent-based modeling; coupling coordination degree



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1. Introduction

Under the combined effects of rapid urbanization, industrial restructuring, and climate change, water-scarce megacities are generally confronted with the compound challenges of tightening water supply–demand relations and accumulating water environmental carrying capacity [1,2]. Population agglomeration, economic expansion, and intensive human activities continuously increase urban water withdrawal and consumption, while the discharge of domestic sewage and industrial wastewater also rises. As a result, water resource scarcity, water environmental pollution, and insufficient ecological water supply become increasingly intertwined [3,4]. For such cities, water system problems are no longer manifested as an imbalance in a single water quantity or water quality factor, but rather as a systemic disorder arising from the coupling between the natural water cycle and the social water cycle [5,6]. The urban water metabolism system describes the processes of water

acquisition, utilization, treatment, discharge, and reuse and can comprehensively reflect the flows, transformations, and feedbacks of water quantity and water quality between natural systems and socioeconomic systems. It therefore provides an important analytical framework for understanding the evolutionary mechanism of urban water systems and supporting sustainable water management [7,8].

A considerable body of research has examined the structural identification, operational diagnosis, and policy simulation of urban water metabolism systems. Existing studies can generally be classified into three categories. The first category consists of comprehensive evaluation methods based on indicator systems, which usually diagnose the status of urban water systems from the perspectives of water resource use efficiency, water environmental carrying capacity, and socioeconomic development level [9–11]. These methods are highly operable and suitable for phased comparison and current-status assessment. However, they mostly rely on static data or limited temporal sections, making it difficult to reveal the dynamic mechanisms through which urban water metabolism systems evolve over time [12–14]. The second category involves system dynamics (SD) models. By constructing feedback relationships among population, economy, resources, and environment, SD models can effectively simulate the long-term evolution of urban water systems under different development scenarios [15–17]. Nevertheless, SD models usually follow a top-down modeling logic and remain limited in representing behavioral heterogeneity among micro-level agents, such as residents and enterprises, as well as the pathways through which policies affect their decisions [18–20]. The third category is agent-based modeling (ABM), which represents adaptive decision making and emergent characteristics in complex systems by defining individual behavioral rules and interaction mechanisms. ABM therefore has distinctive advantages in analyzing behavioral responses and policy effects [21–23]. However, when used alone, ABM often has difficulty systematically representing resource constraints, environmental carrying capacity, and macro-level feedback structures. Overall, although existing studies have made progress in system status evaluation, feedback process simulation, and agent behavior representation, an integrated research framework that simultaneously incorporates micro-level behavioral simulation, macro-level system feedback, and dynamic state identification is still lacking for urban water metabolism systems.

In recent years, the coupling of ABM and SD has increasingly been applied to studies of complex social–economic–environmental systems, providing a new methodological pathway for linking agent behavior with system evolution. For urban water metabolism systems, this coupled modeling approach helps reveal how the behaviors of agents such as governments, residents, and enterprises influence overall system evolution through policy intervention, resource utilization, and pollutant discharge [19,24–28]. However, its application in urban water metabolism assessment remains limited in three respects. First, existing ABM–SD studies mainly emphasize scenario simulation or policy-effect comparison, whereas the coordinated and sustainable state of the water metabolism system is rarely identified explicitly [29,30]. Second, process simulation alone can describe the changing trends of water use, pollutant discharge, and socioeconomic development, but it cannot determine whether the system has entered a coordinated, overloaded, warning, or sustainable state [31–33]. Third, many existing evaluation approaches rely on weighted composite indices, which may weaken the physical interpretability of the results [34–38]. Therefore, this study further introduces a two-dimensional D–PI matrix by combining the coupling coordination degree and the Nemerow index. This matrix links socioeconomic–water environmental coordination with water environmental carrying capacity, allowing the evolutionary process and the operational state of the urban water metabolism system to be interpreted jointly.

Based on this consideration, this study proposes an integrated analytical framework that combines ABM–SD-coupled simulation with D–PI dynamic assessment to characterize the internal mechanism of “micro-level behavior–macro-level feedback–state evolution” in urban water metabolism systems. Specifically, an ABM–SD-coupled model is first constructed to simulate the dynamic evolution of urban water resource utilization, pollutant discharge, and policy regulation through the interactions among government, resident, enterprise, industry, wastewater treatment plant, and environment agents. Then, the coupling coordination degree (D) and the Nemerow index (PI) are introduced to construct a two-dimensional D–PI state space, which enables joint identification of the coordination relationship between socioeconomic development and the water environment and the carrying status of the water environment. This framework thereby achieves an integrated analysis of both the evolutionary process and the coordinated sustainable state of urban water metabolism systems. Taking Beijing as a case study, this paper further identifies the dynamic evolutionary characteristics of its water metabolism system and reveals the non-synchronous evolutionary pattern in which socioeconomic–water environmental coordination improves while carrying pressure accumulates. The findings provide methodological support for dynamic diagnosis, policy optimization, and sustainable management of water systems in water-scarce megacities.

2. Study Area and Data Sources

2.1. Study Area

Beijing is located in the northern part of the North China Plain, between $39^{\circ}28'–41^{\circ}05'$ N and $115^{\circ}20'–117^{\circ}30'$ E. The average elevation is approximately 43.5 m, and the terrain generally descends from the northwest to the southeast, forming a distinct stepped topographic pattern. The interlaced distribution of mountainous and plain areas has an important influence on precipitation convergence and runoff processes. The location of the study area is shown in Figure 1.

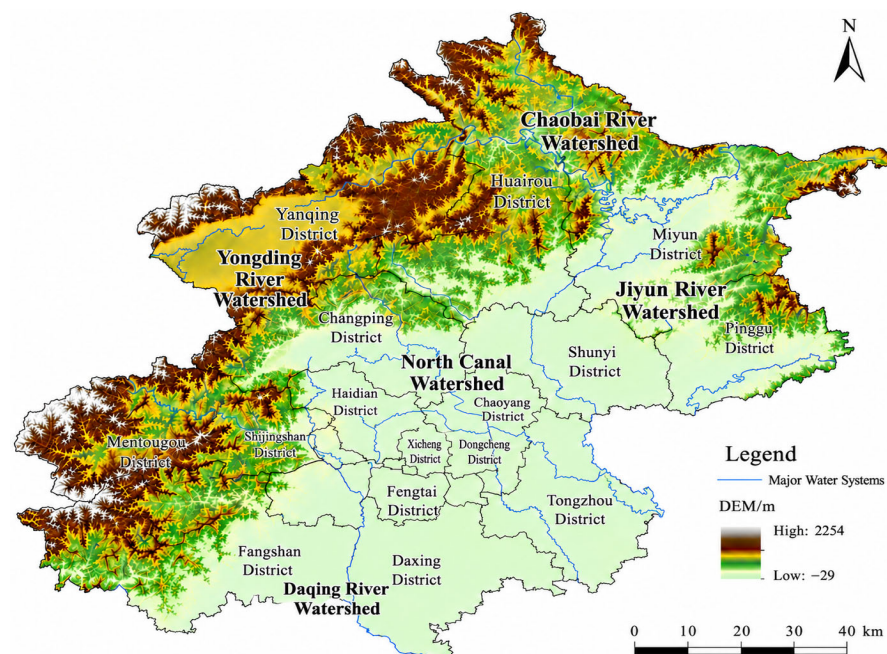


Figure 1. Location of the study area.

Beijing has a warm temperate semi-humid continental monsoon climate, with an annual average temperature of approximately 12.9°C . In terms of precipitation, the multi-year average annual precipitation is approximately 600 mm, but its interannual and seasonal

distribution is highly uneven. Approximately 60% of annual precipitation is concentrated in July and August, indicating a significant imbalance in the spatial and temporal distribution of water resources [39]. The regional water system mainly consists of five major river systems: the Daqing River, Yongding River, Chaobai River, North Canal, and Jiyun River systems. These river systems perform multiple functions, including urban water supply, flood control, and water environmental carrying capacity. Under long-term high-intensity development, utilization, and pollutant discharge, Beijing faces the coexistence of water resource scarcity and water environmental carrying constraints. It is therefore a typical water-scarce megacity in northern China and provides a representative case for studying the urban water metabolism system.

2.2. Data Sources

The data required for calculating the coupling coordination degree model were mainly obtained from the Beijing Statistical Yearbook (2013–2023), the Beijing Statistical Communiqué on National Economic and Social Development (2012–2023), the Beijing Water Affairs Statistical Yearbook (2016–2023), the Beijing Water Resources Bulletin (2012–2023), the Beijing Environmental Statistics Annual Report (2012–2019), the Beijing Statistical Annual Report on Emission Sources (2020–2023), the China Urban Construction Statistical Yearbook (2012–2023), the China Environmental Statistics Annual Report, the Second National Pollution Source Census Bulletin (2017), the Manual of Pollutant Generation and Discharge Accounting Methods and Coefficients for Emission Source Statistical Surveys, the Beijing Design Standard for Sponge City Stormwater Control and Utilization Engineering [40], the Beijing Sponge City Special Plan, and Beijing's 14th Five-Year Plan.

Specifically, data related to the socioeconomic subsystem were mainly derived from the Beijing Statistical Yearbook and the Beijing Statistical Communiqué on National Economic and Social Development. Data related to the water environment subsystem were mainly obtained from the Beijing Water Affairs Statistical Yearbook, the Beijing Water Resources Bulletin, the Beijing Environmental Statistics Annual Report, the Beijing Statistical Annual Report on Emission Sources, and relevant statistical materials on pollution sources and urban construction. The available amount of water resources in Beijing was set as 40% of the total annual water resources. This assumption was adopted because the ratio of water withdrawal to total renewable water resources is commonly used to indicate water resource development intensity, and a threshold of approximately 40% is often regarded as a warning level for high water stress or excessive development intensity in water resource assessment studies. In this study, the 40% threshold was used as a conservative carrying-capacity constraint rather than as the actual exploitable engineering water supply. This treatment helps reflect the ecological and sustainability requirement that socioeconomic water use should not fully occupy renewable water resources. Therefore, the 40% threshold was used as a conservative carrying-capacity constraint rather than as the actual exploitable engineering water supply [41,42].

3. Research Methods

3.1. Research Framework and Technical Route

Considering the complexity of the urban water metabolism system [9,43–45], this study develops a research framework integrating “system simulation–dynamic assessment–evolutionary analysis”. First, under the theoretical framework of complex adaptive systems (CASs), an ABM–SD-coupled simulation model of the urban water metabolism system is constructed. By integrating multi-agent behavioral simulation with the feedback structure of system dynamics, the model characterizes the dynamic evolution of water resource utilization, pollutant discharge, and policy regulation in the urban water system.

Second, a dynamic assessment method centered on the coupling coordination degree (D) and the Nemerow index (PI) is established. On this basis, a two-dimensional D–PI joint identification matrix is further constructed to dynamically identify the sustainable development status of the urban water metabolism system and analyze its evolutionary patterns. The technical route of this study is shown in Figure 2.

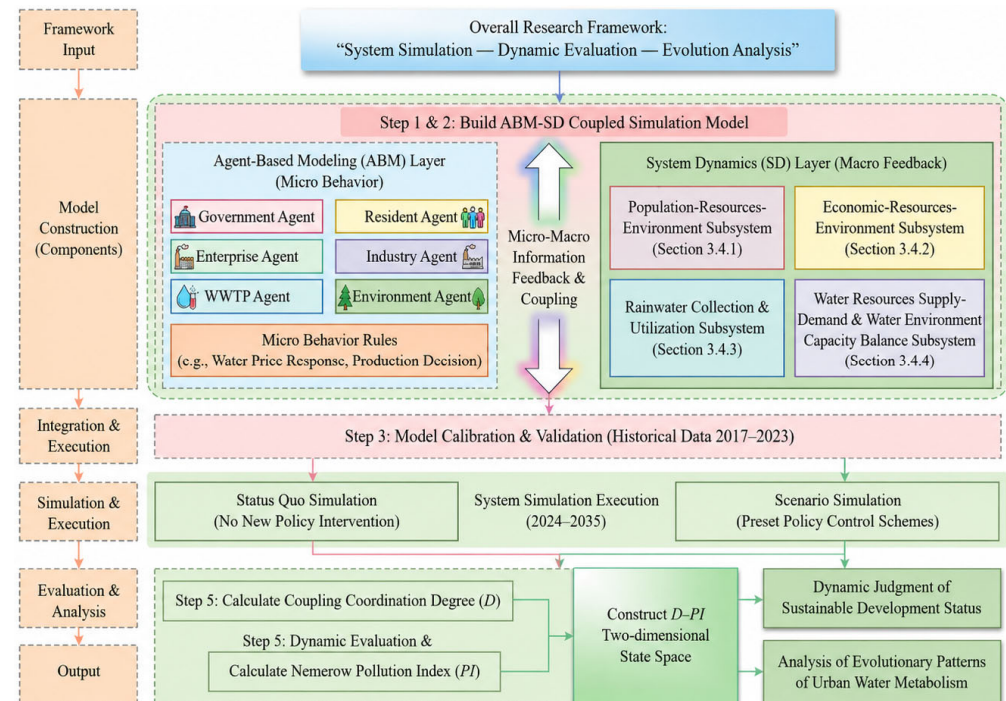


Figure 2. Technical route of this study.

3.2. Overall Framework of the ABM–SD-Coupled Model

Based on complex adaptive systems theory, this study constructs an ABM–SD-coupled simulation model for the urban water metabolism system by integrating micro-level agent behavior with macro-level system feedback, thereby enabling multi-scale simulation of the urban water metabolism system. As shown in Figure 3, in terms of model structure, the ABM layer is used to represent the behavioral decisions of micro-level agents, such as residents and enterprises, in the processes of water resource utilization and pollutant discharge. It also simulates their adaptive responses under policy constraints, including water prices, wastewater treatment fees, and environmental taxes. The SD layer describes the feedback relationships among population, economy, resources, and environment from a macro-level perspective, thereby reflecting the dynamic evolution of water quantity and water quality.

Information exchange between the ABM and SD layers enables the transmission and coupling of micro-level behavior with macro-level system evolution. To further clarify the coupling mechanism, the information exchange between the ABM and SD layers is summarized in Table 1. In the ABM layer, resident agents generate domestic water use, domestic sewage, and corresponding pollutant loads according to water prices, wastewater treatment fees, income level, and basic living demand. Enterprise agents generate industrial water use, wastewater discharge, pollutant loads, and water reuse rates according to production scale, water prices, wastewater treatment fees, environmental taxes, and penalty constraints. These micro-level behavioral outcomes are aggregated and transmitted to the SD layer as input variables for the population–resource–environment subsystem and the economy–resource–environment subsystem. In turn, the SD layer calculates macro-level variables, including total water demand, water resource carrying rate, COD carrying rate,

ammonia nitrogen carrying rate, and the Nemerow index. These outputs are fed back to government and environment agents, forming the basis for subsequent policy regulation. Therefore, the coupling process follows a closed-loop structure of “agent decision making–system aggregation–macro feedback–policy adjustment”. The model includes six types of agents: government, residents, enterprises, industries, wastewater treatment plants, and the environment. Through a closed-loop mechanism of “micro-level behavior–macro-level feedback–policy regulation”, the model simulates the operational process of the urban water metabolism system. This model provides a foundation for the subsequent dynamic assessment of the system.

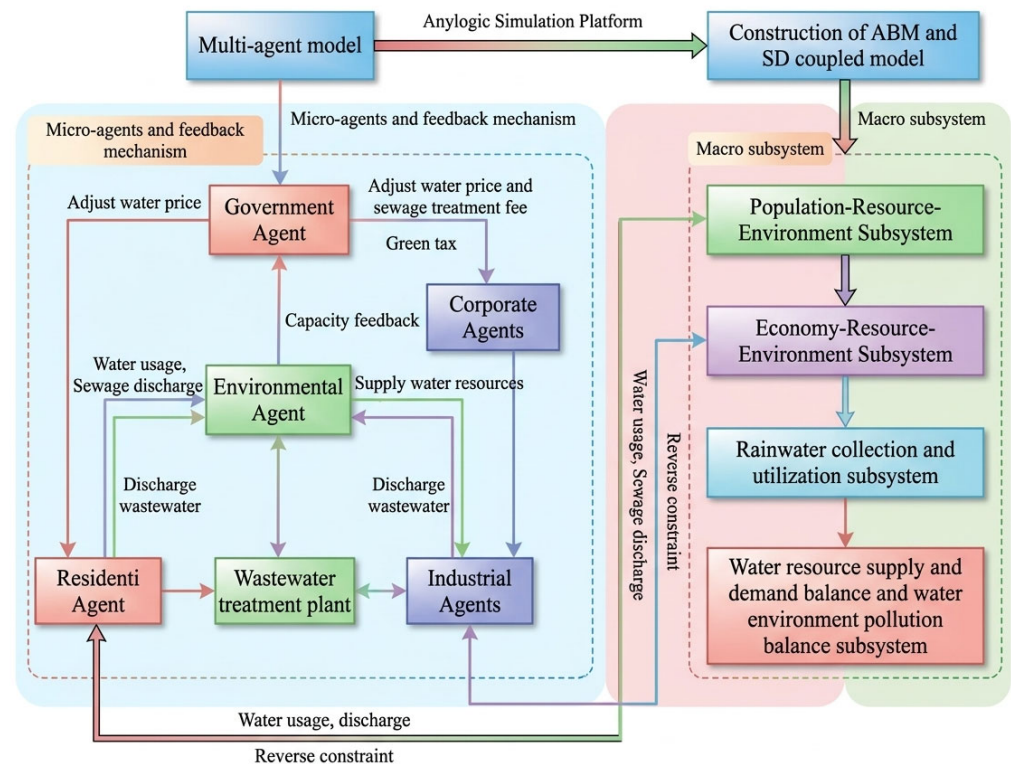


Figure 3. Coupled ABM–SD framework for the urban water metabolism system.

Table 1. Information exchange between the ABM layer and the SD layer.

Direction	Variables Exchanged	Function in the Coupled Model
ABM layer → SD layer	Domestic water use, domestic sewage, domestic COD and ammonia nitrogen loads generated by resident agents	Inputs to the population–resource–environment subsystem
ABM layer → SD layer	Industrial water use, wastewater discharge, COD and ammonia nitrogen loads, and water reuse rates generated by enterprise agents	Inputs to the economy–resource–environment subsystem
ABM layer → SD layer	Wastewater treatment capacity, pollutant removal rates, and reclaimed water supply	Inputs to the water resource supply–demand and water environment carrying capacity balance subsystem

Table 1. *Cont.*

Direction	Variables Exchanged	Function in the Coupled Model
SD layer → ABM layer	Total water demand, water resource carrying rate, COD carrying rate, ammonia nitrogen carrying rate, and PI	Feedback information for government and environment agents
Government/environment feedback → ABM layer	Adjusted water prices, wastewater treatment fees, environmental taxes, penalty intensity, and technology regulation intensity	Policy constraints affecting resident and enterprise behavioral decisions

3.3. Multi-Agent Design and Behavioral Rules

According to the operational characteristics of the urban water metabolism system, six types of agents are defined in the ABM–SD-coupled simulation model: government agents, resident agents, enterprise agents, industry agents, wastewater treatment plant agents, and environment agents. These agents are used to characterize the functional roles and behavioral differences in different actors in water resource utilization, pollutant discharge, and policy regulation [46,47].

A. Government agents. Government agents serve as the core regulators of the system and are mainly responsible for formulating water resource and environmental management policies. By adjusting policy instruments such as water prices, wastewater treatment fees, and environmental taxes, government agents guide residents and enterprises to change their water use and pollutant discharge behaviors. When the system detects an imbalance between water resource supply and demand or pollutant discharge exceeding environmental capacity, government agents increase the intensity of policy regulation.

B. Resident agents. Resident agents consume water resources through daily domestic activities and generate domestic sewage. In the model, resident agents are assumed to have bounded rationality and to adjust their water use according to both basic living demand and water-related expenditure. For each resident household, monthly water expenditure is calculated according to the tiered residential water price and wastewater treatment fee. When the ratio of water-related expenditure to disposable income exceeds the assumed affordability threshold, the agent reduces its water consumption in the next simulation period. If the proportion of resident agents exceeding the affordability threshold reaches a preset warning level, this information is fed back to the government agent, which then adjusts the intensity of water price regulation. Through this mechanism, household-level responses to price changes are aggregated into urban domestic water demand and domestic pollutant discharge in the SD layer.

C. Enterprise agents. Enterprise agents consume water resources and discharge industrial wastewater during production activities. In the model, enterprise agents are assumed to make water use and pollutant discharge decisions according to the principle of cost minimization under environmental regulation. The total cost of each enterprise includes the water supply cost, wastewater treatment fee, environmental tax, wastewater reuse or pretreatment cost, and potential penalties for exceeding discharge standards. Under changes in water prices, wastewater treatment fees, environmental taxes, or penalty intensity, enterprise agents compare the costs of alternative production and discharge strategies and select the option with the lowest total cost. The selected strategy determines industrial water use, wastewater discharge, pollutant loads, and the industrial water reuse rate. These outcomes are then aggregated and transmitted to the SD layer to update industrial water demand and pollutant emissions.

D. Industry agents. Industry agents represent changes in the macroeconomic structure. In the model, the economic system is divided into the primary, secondary, and tertiary

industries, among which the secondary industry serves as an important link between micro-level enterprise behavior and the macroeconomic system.

E. Wastewater treatment plant agents. Wastewater treatment plant agents are responsible for the centralized treatment of domestic sewage and industrial wastewater. Through pollutant reduction processes, they reduce the pollution load entering the water environment.

F. Environment agents. Environment agents provide water resources to the system and perform the function of pollutant assimilation. When water resource consumption or pollutant discharge exceeds environmental carrying capacity, environment agents feed back system status information to government agents.

The behavioral outcomes of all agents are aggregated in the ABM layer and then used as input variables for the system dynamics model, thereby driving the overall evolution of the urban water metabolism system. This coupling is necessary because a conventional SD model can describe macro-level feedbacks, but cannot explicitly capture heterogeneous behavioral responses to policy instruments. In this study, ABM is used to simulate how residents and enterprises adjust water use, wastewater discharge, and reuse behavior in response to water prices, wastewater treatment fees, and environmental taxes. These agent-level responses are then aggregated into the SD layer, forming a closed loop of “policy signal–agent response–system feedback”.

3.4. Construction of the ABM–SD-Coupled Model

The ABM–SD-coupled simulation model of the urban water metabolism system constructed in this study consists of four subsystems: the population–resource–environment subsystem, the economy–resource–environment subsystem, the rainwater collection and utilization subsystem, and the water resource supply–demand balance and water environmental carrying capacity balance subsystem. Together, these subsystems are used to systematically characterize the evolution of water quantity and water quality in the urban water metabolism process. The information flows among the four subsystems are organized around water demand, water supply, and pollutant loads. The population subsystem calculates domestic water use, sewage generation, and domestic COD and ammonia nitrogen loads. The economy subsystem calculates industrial and service-sector water use, wastewater generation, pollutant emissions, and reclaimed water reuse. The rainwater subsystem estimates runoff reuse and non-point source pollution loads. These outputs are finally aggregated in the supply–demand and environmental capacity balance subsystem to evaluate water resource pressure, pollutant assimilation pressure, and the overall carrying status.

3.4.1. Population–Resource–Environment Subsystem

In this study, Beijing’s permanent resident population is divided into urban residents and rural residents, both of which are incorporated into resident agents. Within the constructed system, the model simulates the water quantity and water quality metabolism process of domestic water use, including water withdrawal, water consumption, transformation into domestic sewage, treatment, and discharge.

The subsystem focuses on representing the influence of water prices on the water use and pollutant discharge behaviors of metered urban households, while parameters related to rural residents are determined using the pollutant generation and discharge coefficient method. The structure of the population–resource–environment subsystem is shown in Figure 4.

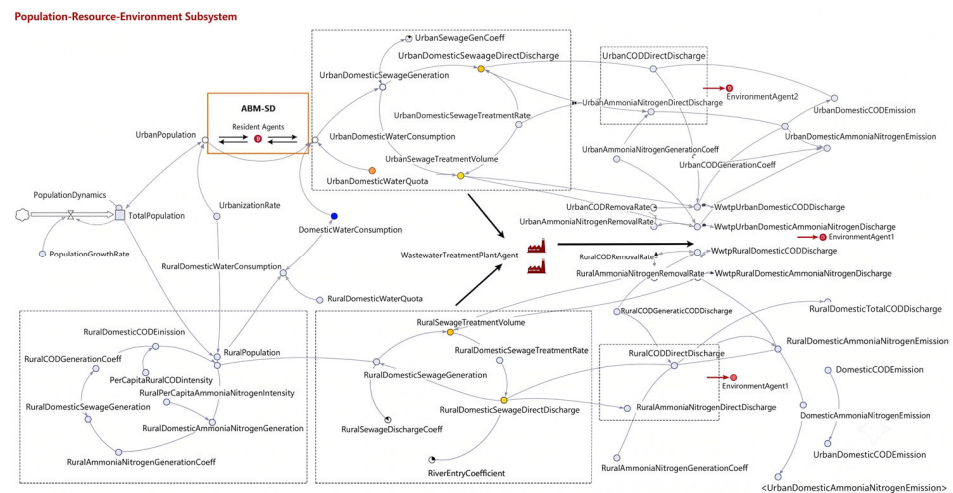


Figure 4. Flowchart of the population–resource–environment subsystem.

3.4.2. Economy–Resource–Environment Subsystem

This subsystem characterizes the process through which industrial production activities create economic value while simultaneously causing resource consumption and environmental pollution. In the model, industry agents are divided into primary, secondary, and tertiary industries to simulate the water quantity–water quality metabolism process of industrial water withdrawal, wastewater generation, wastewater treatment, and discharge.

Wastewater treatment plant agents are introduced into this subsystem to conduct centralized treatment of industrial wastewater. Part of the treated wastewater is reused as reclaimed water after advanced treatment, while the remaining effluent is discharged into the environment. On this basis, the total emissions of chemical oxygen demand (COD) and ammonia nitrogen are calculated. The structure of this subsystem is shown in Figure 5.

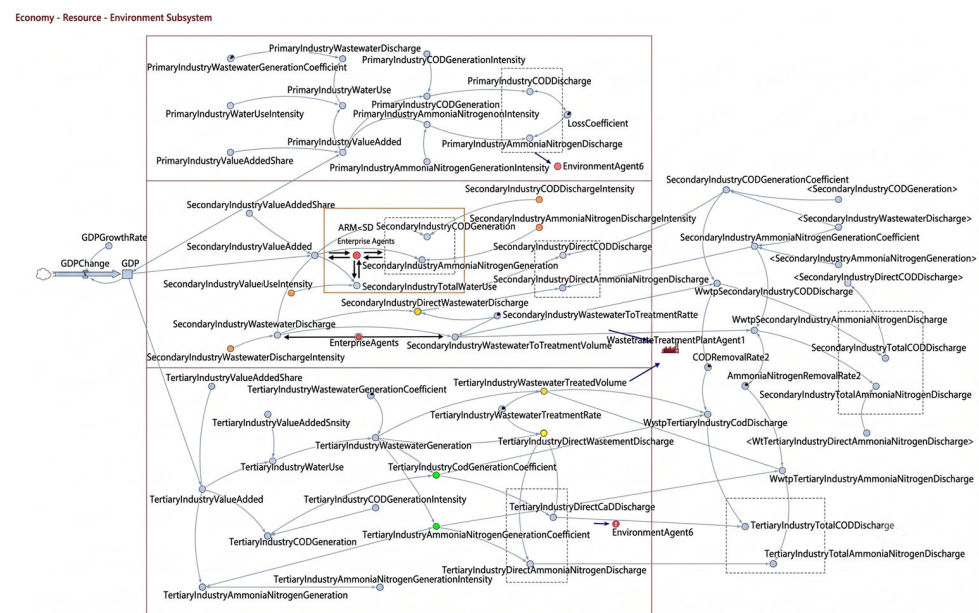


Figure 5. Flowchart of the economy–resource–environment subsystem.

3.4.3. Rainwater Collection and Utilization Subsystem

This subsystem mainly involves the collection, conveyance, purification, and reuse of surface runoff. Runoff collected through the rainwater pipe network is partly discharged into rivers, lakes, and reservoirs, and partly treated and reused for irrigation, groundwater recharge, landscape greening, and municipal water use, as shown in Figure 6.

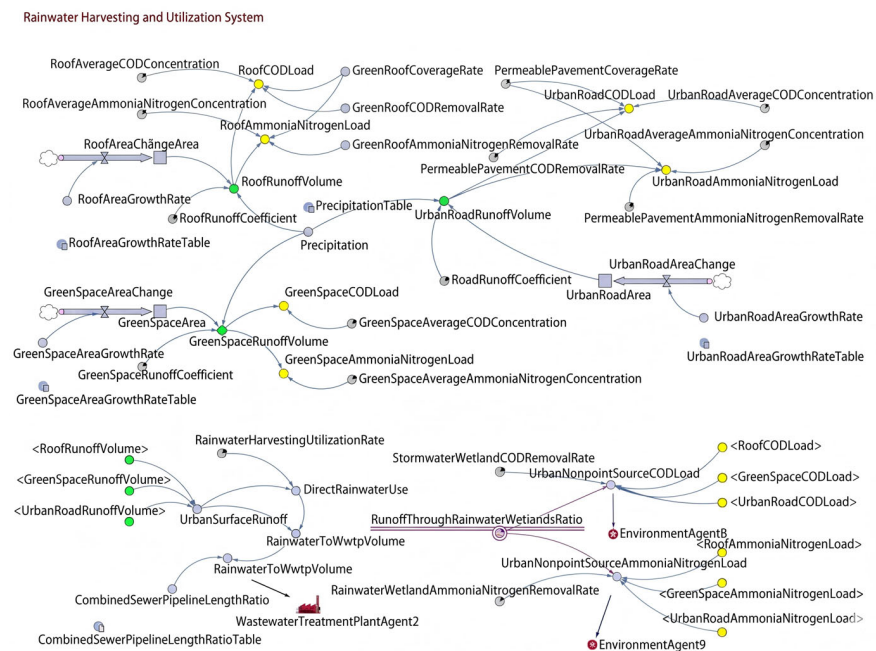


Figure 6. Flowchart of the rainwater collection and utilization subsystem.

Based on Beijing's land use types, the Beijing Urban and Rural Planning Land Use Classification Standard, and the Beijing Design Standard for Sponge City Stormwater Management and Utilization Engineering, the underlying surface of Beijing is divided into three categories: rooftops, green spaces, and urban roads. Corresponding runoff coefficients are selected for different underlying surfaces, and the average concentrations of runoff pollutants are combined to estimate the urban non-point source pollution load.

3.4.4. Water Resource Supply–Demand Balance and Water Environmental Carrying Capacity Balance Subsystem

The water resource supply–demand balance and water environmental carrying capacity balance subsystem characterizes the carrying status of the urban water metabolism system from the perspectives of water quantity and water quality, respectively.

In terms of water quantity, the ABM–SD-coupled simulation model is used to calculate domestic water use, water use by the three industries, and ecological environmental water use. These water demands are then compared with the total available water supply, including local water resources, transferred water, rainwater utilization, and reclaimed water.

In terms of water quality, the model calculates domestic pollutant loads, production-related pollutant loads, and non-point source pollution loads and compares them with water environmental carrying capacity. When the water resource carrying index or the water environmental carrying index is greater than 1, it indicates an imbalance between water resource supply and demand or an overload of water environmental pollutant assimilation capacity, respectively. In such cases, regulation should be implemented by optimizing the socioeconomic structure, strengthening water environmental technical measures, and adjusting water environmental economic policies. When both carrying indices are less than 1, the urban water metabolism system and socioeconomic development are considered to be in a coordinated state. The subsystem structure is shown in Figure 7.

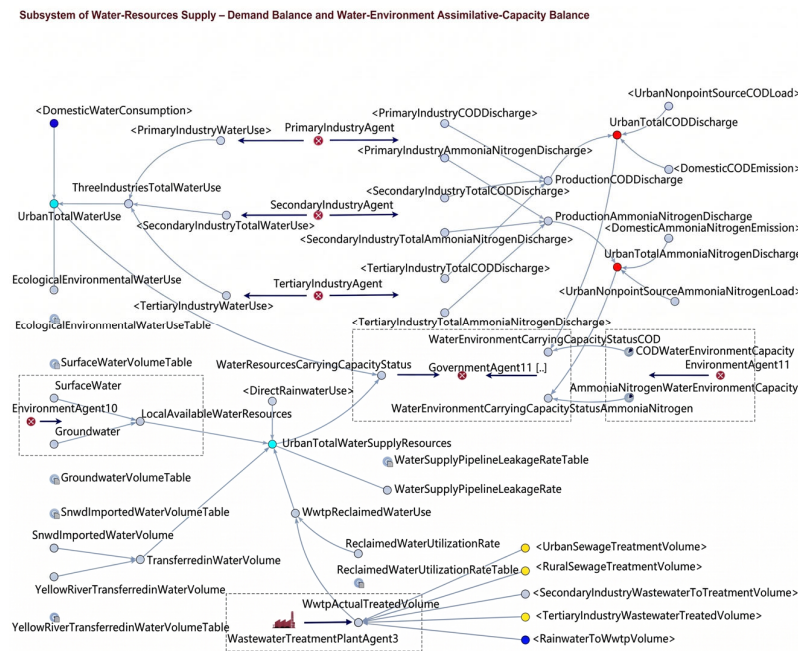


Figure 7. Flowchart of the water resource supply–demand balance and water environmental carrying capacity balance subsystem.

3.5. D–PI Dynamic Assessment Method

To improve the readability of the equations, the main symbols used in the dynamic assessment method are summarized in Table 2.

Table 2. Main symbols used in the dynamic assessment equations.

Symbol	Meaning
$s(x)$	Integrated efficacy value of the socioeconomic subsystem
$e(y)$	Integrated efficacy value of the water environment subsystem
C	Coupling degree
D	Coupling coordination degree
T	Comprehensive coordination index
α, β	Contribution coefficients of the socioeconomic and water environment subsystems
PI	Nemerow index
C_{imax}	Maximum value among the water resource carrying rate, COD carrying rate, and ammonia nitrogen carrying rate
C_{iave}	Average value of the three carrying rates

3.5.1. Calculation Method of the Coupling Coordination Degree (D)

The coupling coordination degree model is used to measure the coordinated development level between the socioeconomic subsystem and the water environment subsystem [48–50]. First, the integrated efficacy values of the socioeconomic subsystem and the water environment subsystem are calculated separately. Then, the coupling degree and coupling coordination degree of the two subsystems are calculated to reflect the internal coordination status of the system.

For the socioeconomic subsystem, assuming that there are m indicators, its integrated efficacy value is calculated using Equation (1).

$$s(x) = \sqrt[n_s]{d_{s1} \cdot d_{s2} \cdots d_{s_{n_s}}} \quad (1)$$

For the water environment subsystem, assuming that there are n indicators, its integrated efficacy value is calculated using Equation (2).

$$e(y) = \sqrt[n_e]{d_{e1} \cdot d_{e2} \cdots d_{en_e}} \quad (2)$$

where $s(x)$ and $e(y)$ denote the integrated efficacy values of the socioeconomic subsystem and the water environment subsystem, respectively; $d_{s1}, d_{s2}, \dots, d_{sn_s}$ denote the standardized efficacy coefficients of the indicators in the socioeconomic subsystem; $d_{e1}, d_{e2}, \dots, d_{en_e}$ denote the standardized efficacy coefficients of the indicators in the water environment subsystem; and n_s and n_e denote the number of indicators in the two subsystems, respectively. The integrated efficacy values $s(x)$ and $e(y)$ are calculated using the geometric mean of the standardized efficacy coefficients and are then used as inputs to the coupling coordination degree model.

Taking the integrated efficacy values $s(x)$ and $e(y)$ as the evaluation indicators of the two subsystems, the coupling degree C and the coupling coordination degree D are calculated.

The coupling degree C is calculated using Equation (3).

$$C = 2\sqrt{\frac{s(x) \cdot e(y)}{[s(x) + e(y)]^2}} \quad (3)$$

where C ranges from 0 to 1. A larger value of C indicates a higher degree of coupling between the two subsystems.

The coupling coordination degree D is calculated using Equations (4) and (5).

$$D = \sqrt{TC} \quad (4)$$

$$T = \alpha \cdot s(x) + \beta \cdot e(x) \quad (5)$$

where D represents the coupling coordination degree. Higher D values indicate greater system coordination. T is the comprehensive coordination index of the socioeconomic subsystem and the water environment subsystem. α and β are undetermined coefficients representing the relative importance and contribution of the socioeconomic subsystem and the water environment subsystem to the overall system, respectively, with $\alpha + \beta = 1$ and β . Considering that the two subsystems are equally important to the urban water metabolism system and have no priority in terms of contribution, both α and β are assigned a value of 0.5 in this study [51,52]. The evaluation criteria are shown in Table 3.

Table 3. Evaluation criteria for the coupling coordination status of the urban water metabolism system.

General Category	Coupling Coordination Degree D	Subcategory	Comparison of Efficacy Values	Basic Type of Coupling Coordination Development
Uncoupled	$0 < D \leq 0.4$	Severe disorder and recession	$s(x) < e(y)$	Severe disorder and recession with socioeconomic impairment
			$s(x) = e(y)$	Severe disorder and recession with both socioeconomic and water environment impairment
			$s(x) > e(y)$	Severe disorder and recession with water environment impairment
	$0.4 < D \leq 0.5$	Disorder and recession	$s(x) < e(y)$	Disorder and recession with socioeconomic impairment
			$s(x) = e(y)$	Disorder and recession with antagonism between socioeconomic development and the water environment
			$s(x) > e(y)$	Disorder and recession with water environment impairment

Table 3. Cont.

General Category	Coupling Coordination Degree D	Subcategory	Comparison of Efficacy Values	Basic Type of Coupling Coordination Development
Uncoupled	$0.5 < D \leq 0.6$	Barely coordinated development	$s(x) < e(y)$	Barely coordinated development with socioeconomic lag
			$s(x) = e(y)$	Barely coordinated development with adjustment between socioeconomic development and the water environment
			$s(x) > e(y)$	Barely coordinated development with water environment lag
Coupled	$0.6 < D \leq 0.7$	Primary coordinated development	$s(x) < e(y)$	Primary coordinated development with socioeconomic lag
			$s(x) = e(y)$	Primary coordinated development with synchronization between socioeconomic development and the water environment
			$s(x) > e(y)$	Primary coordinated development with water environment lag
	$0.7 < D \leq 0.8$	Moderate coordinated development	$s(x) < e(y)$	Moderate coordinated development with socioeconomic lag
			$s(x) = e(y)$	Moderate coordinated development with synchronization between socioeconomic development and the water environment
			$s(x) > e(y)$	Moderate coordinated development with water environment lag
	$0.8 < D \leq 1$	Well-coordinated development	$s(x) < e(y)$	Well-coordinated development with socioeconomic lag
			$s(x) = e(y)$	Well-coordinated development with synchronization between socioeconomic development and the water environment
			$s(x) > e(y)$	Well-coordinated development with water environment lag

3.5.2. Calculation Method of the Water Environmental Carrying Status (PI)

To identify the carrying status of the urban water metabolism system under the combined effects of water resource utilization and pollutant discharge, this study introduces the Nemerow index (PI) to comprehensively evaluate the water environmental carrying status. In this study, water environmental carrying status refers to the integrated pressure state under both water resource and pollutant assimilation constraints. The index integrates three indicators: the water resource carrying rate, the COD carrying rate, and the ammonia nitrogen carrying rate. These indicators were selected because they represent the main water quantity and water quality constraints in Beijing's urban water metabolism system. Other indicators, such as total nitrogen and total phosphorus, were not included because consistent long-term data and corresponding environmental capacity parameters were not sufficiently available for the whole simulation period. It can effectively reflect the overall pressure level of the water environment system [53–56]. The Nemerow index is calculated as follows:

$$PI = \sqrt{\frac{C_{imax}^2 + C_{iave}^2}{2}} \quad (6)$$

where PI denotes the Nemerow index used to evaluate the water environmental carrying status of the study area; C_{imax} is the maximum value among the water resource carrying rate, COD carrying rate, and ammonia nitrogen carrying rate; and C_{iave} is the average value of these three carrying rates.

Based on existing studies and the actual characteristics of the urban water metabolism system, the water environmental carrying status is divided into four levels: far from

overload, critical overload, slight overload, and severe overload. The specific evaluation criteria are shown in Table 4.

Table 4. Evaluation criteria for the water environmental carrying status of the urban water metabolism system.

Nemerow Index PI	$0 \leq PI < 0.5$	$0.5 \leq PI < 1$	$1 \leq PI < 1.5$	$PI \geq 1.5$
Carrying status	Far from overload	Critical overload	Slight overload	Severe overload

3.5.3. Two-Dimensional D–PI Joint Identification Method

Based on the calculated results of the coupling coordination degree (D) and the Nemerow index (PI), a two-dimensional D–PI joint identification matrix is further constructed to identify the coordinated and sustainable development status of the urban water metabolism system. In this matrix, the coupling coordination degree D is used as the horizontal dimension, while the Nemerow index PI is used as the vertical dimension. The system status is divided into four categories: the “Disorder–Overload Zone”, the “Pressure–Adjustment Zone”, the “Warning–Optimization Zone”, and the “Coordination–Sustainability Zone”.

The classification rules are as follows.

(1) Red zone. This zone refers to cases where the water environmental carrying status is severely overloaded ($PI \geq 1.5$) and the coupling coordination degree is very low ($D \leq 0.6$), or where the water environmental carrying status is slightly overloaded ($1 \leq PI < 1.5$) and the coupling coordination degree is no higher than 0.5. In this zone, the water environmental carrying status has reached a dangerous overload level, while the system’s coordination capacity is also weak. The urban water metabolism system may enter a state of disorder and decline, requiring urgent intervention.

(2) Orange zone. This zone includes cases where the water environmental carrying status is severely overloaded but the coordination degree is relatively high ($D > 0.6$), where the water environmental carrying status is slightly overloaded but the coordination degree is greater than 0.5, or where the system is in a critical overload state and the coordination degree is no higher than 0.5. This indicates that although the system has a certain coordination capacity, it still faces substantial carrying pressure and remains at a relatively high risk level. Effective measures are therefore needed to reduce the overload status and prevent further deterioration.

(3) Yellow zone. This zone includes cases where the water environmental carrying status is at a critical overload level and the coordination degree is greater than 0.6, where the water environmental carrying status is slightly overloaded and the coordination degree is greater than 0.7, or where the system is far from overload but the coordination degree is no higher than 0.6. This indicates that the system has a certain coordination capacity, but its carrying pressure is approaching a critical level. Measures should be taken to continuously improve the water environment and guide the system back toward a coordinated and sustainable development status.

(4) Green zone. This zone indicates a favorable water environmental carrying status ($PI < 1$) and a relatively high coupling coordination degree ($D \geq 0.7$). In this case, the water environment is sufficient to support current socioeconomic activities, and socioeconomic development and the water environment are developing in a coordinated manner. This represents a relatively satisfactory system status.

Based on the above classification rules, the dynamic evolution status of the urban water metabolism system in different years can be comprehensively identified. The specific classification matrix is shown in Figure 8.

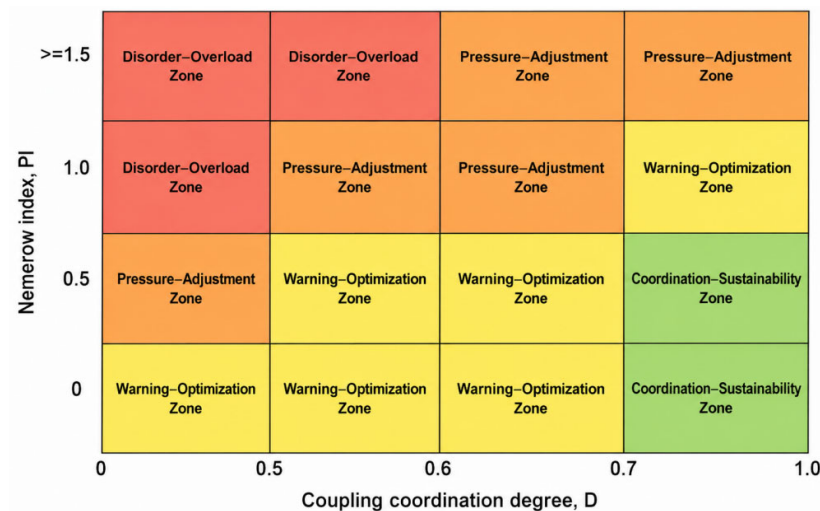


Figure 8. Classification matrix of the coordinated and sustainable development level of the urban water metabolism system.

4. Results and Analysis

4.1. Model Validation

Model validation was conducted by comparing the model output data with the corresponding historical data. In this study, error analysis was used to evaluate the validity of the model. The relative error was calculated as follows:

$$M_t = \frac{S_t - H_t}{H_t} \quad (7)$$

where M_t represents the relative error in year t , S_t represents the simulated value in year t , and H_t represents the historical value in year t .

The model takes 2017 as the base year and runs through 2035. The simulated values from 2017 to 2023 were compared with the observed historical values for historical validation. Total population, gross domestic product (GDP), total water use, total chemical oxygen demand (COD) emissions, and total ammonia nitrogen emissions were selected as validation variables. These variables were selected because they represent the main socioeconomic, water resource, and water environmental outputs of the model. Total population and GDP reflect socioeconomic development; total water use represents water resource utilization, and COD and ammonia nitrogen emissions reflect water environmental pressure. Therefore, they can provide a comprehensive validation of the key outputs of the urban water metabolism model. Selected model parameter values are presented in Table 5, and the validation results are shown in Table 6 and Figure 9.

Table 5. Selected parameter values of the model.

Parameter	Unit	2017	2018	2019	2020	2021	2022	2023
Urbanization rate	%	86.9	87.1	87.4	87.5	87.5	87.6	87.8
Population growth rate	%	−0.45	−1.23	−0.73	−0.50	−0.18	−1.96	0.7
GDP growth rate	%	6.8	6.7	6.1	1.1	8.8	0.7	5.2
Proportion of value added of the primary industry	%	0.4	0.4	0.3	0.3	0.3	0.3	0.2
Proportion of value added of the secondary industry	%	16.9	16.5	16.0	16.0	18.0	15.9	14.9

Table 5. Cont.

Parameter	Unit	2017	2018	2019	2020	2021	2022	2023
Proportion of value added of the tertiary industry	%	82.7	83.1	83.7	83.7	81.7	83.9	84.8
Water use intensity of the primary industry	m ³ /CNY 10,000	415.91	346.60	322.55	298.25	252.65	233.75	237.91
Water use intensity of the tertiary industry	m ³ /CNY 10,000	2.35	2.18	2.07	1.73	1.79	1.68	1.68
COD generation intensity of the primary industry	kg COD/CNY 10,000	7.22	2.99	2.41	10.52	11.65	10.56	11.87
Ammonia nitrogen generation intensity of the primary industry	kg NH ₃ -N/CNY 10,000	0.062	0.040	0.027	0.150	0.181	0.168	0.166
COD generation intensity of the tertiary industry	kg COD/CNY 10,000	0.099	0.073	0.051	0.047	0.038	0.033	0.106
Ammonia nitrogen generation intensity of the tertiary industry	kg NH ₃ -N/CNY 10,000	0.0079	0.0051	0.0036	0.003	0.0022	0.0019	0.0096
Urban sewage treatment rate	%	95.0	95.1	96.2	96.6	97.2	98.1	98.2
Rural sewage treatment rate	%	45.0	62.6	67.3	72.4	74.6	83.5	84.9
COD removal rate of wastewater treatment plants	%	87	87	87	87	87	87	87
Ammonia nitrogen removal rate of wastewater treatment plants	%	90	90	90	90	90	90	90
Reclaimed water reuse rate	%	60.7	56.5	57.6	61.9	55.6	54.2	55.5
South-to-North Water Diversion inflow	10 ⁸ m ³	10.77	11.92	9.85	8.82	12.51	11.07	9.80
Yellow River water diversion inflow	10 ⁸ m ³	0	0	0.92	0.52	0.84	1.03	0.80
Ecological environmental water use	10 ⁸ m ³	12.6	13.4	16.0	17.4	16.7	16.4	16.4
Precipitation	mm	592	590	506	560	924	482	727
Leakage rate of water supply pipelines	%	11.60	10.86	9.32	9.85	9.79	9.30	8.80
Proportion of combined sewer pipelines	—	0.080	0.068	0.068	0.060	0.056	0.052	0.048

Note: (1) CNY denotes Chinese yuan; COD denotes chemical oxygen demand. (2) Urban and rural sewage treatment rates refer to the proportions of sewage effectively treated in urban and rural areas, respectively. The COD and ammonia nitrogen removal rates of wastewater treatment plants represent the average pollutant removal efficiencies used in the model. The reclaimed water reuse rate was calculated as the proportion of reclaimed water reuse relative to treated wastewater. Precipitation refers to the annual precipitation in Beijing. These parameters were obtained from statistical yearbooks, water resources bulletins, environmental statistics, and relevant planning documents.

Table 6. Comparison of simulated and observed values for model validation.

Variable	Type	2017	2018	2019	2020	2021	2022	2023
Total population, 10,000 persons	Simulated value	2195.56	2193.30	2190.67	2189.14	2188.05	2187.61	2183.25
	Observed value	2194.40	2191.70	2190.10	2189.00	2188.60	2184.30	2185.80
	Error	0.05%	0.07%	0.03%	0.01%	−0.03%	0.15%	−0.12%
GDP, CNY 100 million	Simulated value	29,984.94	33,192.03	35,492.03	37,723.90	38,255.75	41,765.56	42,279.40
	Observed value	29,883.00	33,106.00	35,445.10	35,943.30	41,045.60	41,540.90	43,760.70
	Error	0.34%	0.26%	0.13%	4.95%	−6.80%	0.54%	−3.39%
Total water use, 10 ⁸ m ³	Simulated value	37.39	37.82	39.55	39.23	39.13	38.52	38.60
	Observed value	39.50	39.30	41.70	40.60	40.80	40.00	40.70
	Error	−5.34%	−3.76%	−5.15%	−3.37%	−4.10%	−3.69%	−5.16%
Total COD emissions, t COD	Simulated value	263,337.05	182,192.33	136,170.98	143,607.56	144,451.10	125,359.7	117,937.23
	Observed value	263,391.92	184,972.02	140,866.90	156,814.23	136,388.90	125,605.70	118,521.79
	Error	−0.02%	−1.50%	−3.33%	−8.42%	5.91%	−0.20%	−0.49%
Total ammonia nitrogen emissions, t NH ₃ -N	Simulated value	31,318.24	19,624.30	14,642.57	13,542.50	12,108.15	10,218.09	9687.24
	Observed value	31,301.86	19,876.25	15,175.72	14,687.93	11,170.15	10,362.55	9724.67
	Error	0.05%	−1.27%	−3.51%	−7.80%	8.40%	−1.39%	−0.38%

In system dynamics model validation, relative errors within approximately 10% are commonly considered acceptable for medium- and long-term simulation models, particularly when the model is used to reproduce the overall evolutionary trend rather than exact point-by-point prediction. Therefore, the relative error threshold of 10% was adopted in this study to evaluate whether the model can reasonably reproduce the historical evolution of Beijing's urban water metabolism system. The historical validation results show that the absolute values of the simulation errors for the selected variables are all controlled within 10%, indicating that the model falls within the acceptable accuracy range. Therefore, the urban water metabolism simulation model for Beijing passes the validity test. The relatively larger deviations in some years can be mainly attributed to abrupt socioeconomic fluctuations and changes in statistical or management conditions. For example, GDP showed larger errors in 2020 and 2021, which may be related to the short-term disturbance caused by the COVID-19 pandemic and the subsequent economic recovery. The errors in total water use were relatively stable but generally negative, suggesting that the model slightly underestimated actual water demand, possibly because temporary changes in ecological water replenishment and interannual precipitation were not fully captured. For COD and ammonia nitrogen emissions, the larger deviations in 2020 and 2021 may be associated with changes in domestic sewage generation, wastewater treatment operation, and statistical accounting during the same period. Overall, these deviations did not affect the model's ability to reproduce the main evolutionary trends of Beijing's urban water

metabolism system. The simulation results can reasonably reflect the actual conditions of Beijing and can be further used for scenario simulation and analysis by adjusting relevant parameters. To test the behavioral rationality of the model in response to price-based policies, sensitivity analyses of water prices, wastewater treatment fees, and environmental taxes were further conducted.

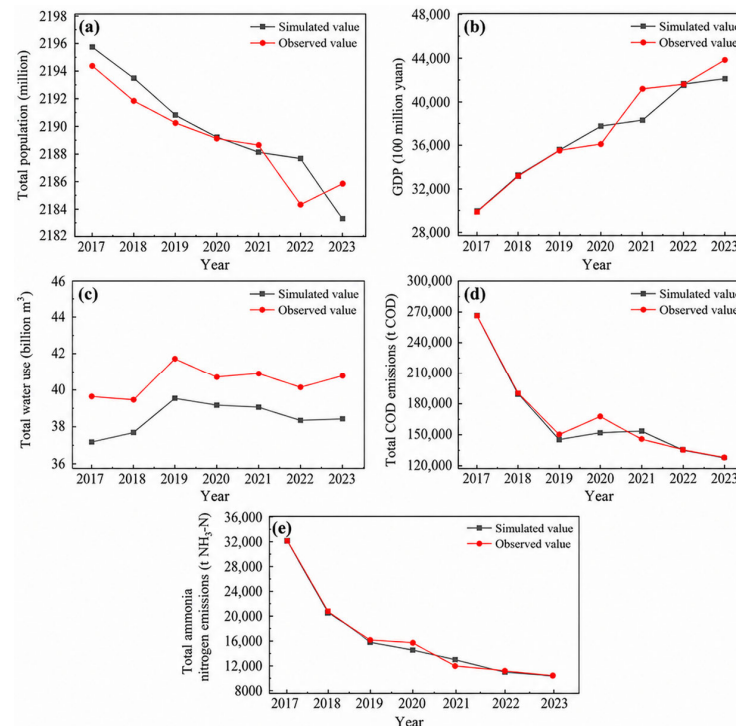


Figure 9. Model validation results: (a) total population; (b) GDP; (c) total water use; (d) total COD emissions; and (e) total ammonia nitrogen emissions.

At the resident level, as shown in Figure 10a,b, residential water use decreases significantly with the increase in tiered water prices, indicating that water prices have a strong regulatory effect on residents' water use behavior. In contrast, the increase in wastewater treatment fees has a relatively weak effect on water use, suggesting that residents are more sensitive to water prices than to wastewater treatment fees. This is mainly because water charges account for a larger proportion of household water-related expenditures, and tiered water pricing has a progressive structure, thereby strengthening its price constraint effect.

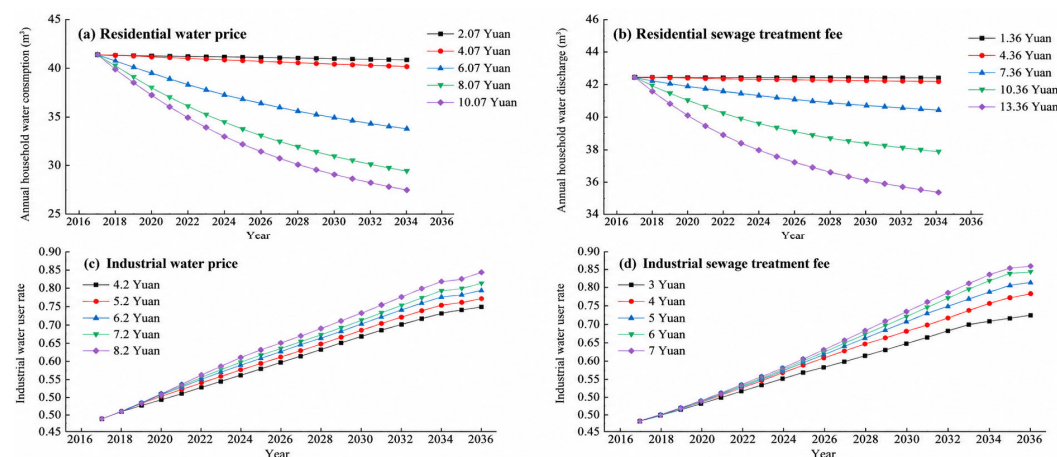


Figure 10. Results of the price sensitivity test for water prices, wastewater treatment fees, and environmental tax policies.

At the enterprise level, as shown in Figure 10c,d, increases in both industrial water prices and wastewater treatment fees can effectively promote improvements in the water reuse rate of enterprises, with wastewater treatment fees exerting a more pronounced effect. This indicates that, compared with water withdrawal costs, pollutant discharge costs impose stronger constraints on enterprise behavior and can more directly drive enterprises to adopt water-saving and emission-reduction measures. In addition, as price levels increase, the improvement in enterprise water reuse rates gradually slows down, showing a certain diminishing marginal effect.

Overall, price-based policies play a significant role in regulating the urban water metabolism system, but different policy instruments operate through different pathways. Water prices mainly affect residential water use behavior, whereas wastewater treatment fees have a more prominent guiding effect on water saving and emission reduction by enterprises. This suggests that, in practical water resource management, differentiated policy combinations should be implemented for different agents to achieve the coordinated optimization of water resource use efficiency and water environmental protection. The sensitivity results also suggest that environmental taxes alone have a relatively limited regulatory effect on enterprise behavior. Therefore, tax-based instruments should be combined with stricter discharge standards, penalty mechanisms, and technological upgrading measures to strengthen emission-reduction incentives.

4.2. Analysis of Simulation Results of the Water Metabolism System

On the basis of model construction and parameter calibration, Beijing's urban water metabolism system was simulated according to historical development trends and current water resource management policies. The simulation period was set from 2017 to 2035, with a time step of one year. The period from 2017 to 2023 was used for model calibration and validation, while the period from 2024 to 2035 was used as the simulation output stage to characterize the dynamic evolutionary characteristics of the urban water metabolism system and provide time-series data support for subsequent dynamic assessment.

4.2.1. Evolutionary Characteristics of Population and Economic Development

As shown in Figures 11 and 12, population and economic development in Beijing show significant structural evolutionary characteristics during the study period. The total population grows slowly, increasing from approximately 21.94 million in 2017 to approximately 22.30 million in 2035, indicating a relatively limited overall growth rate. However, the population structure changes markedly: the urban population continues to increase, while the rural population gradually decreases, reflecting the continued advancement of urbanization.

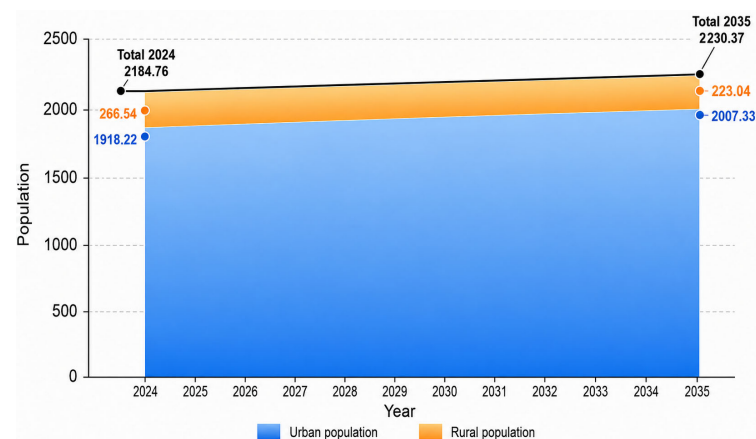


Figure 11. Simulated population changes in Beijing from 2024 to 2035.

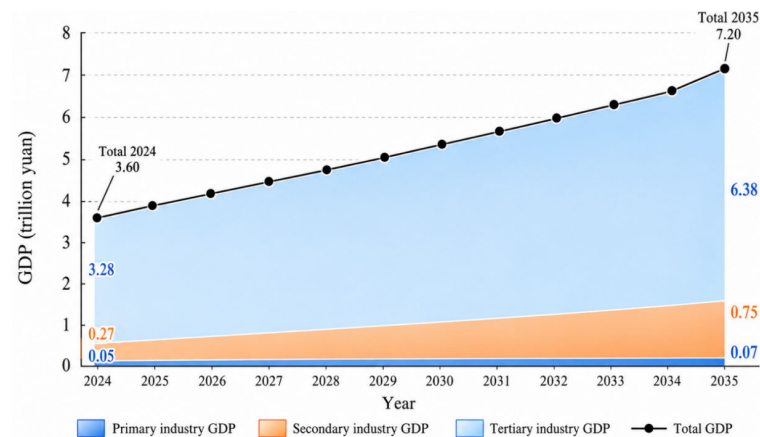


Figure 12. Simulated GDP changes in Beijing from 2024 to 2035.

Meanwhile, Beijing's total economic output increases steadily, and GDP shows a continuous growth trend. The proportion of the tertiary industry continues to rise and becomes the dominant driver of economic growth, whereas the proportion of the primary industry continues to decline and the secondary industry remains relatively stable. Population urbanization and industrial structure upgrading jointly drive changes in the structure of urban water demand and pollutant discharge, providing the fundamental driving forces for subsequent changes in water resource utilization and environmental pressure.

4.2.2. Changes in Water Resource Utilization Structure

As shown in Figure 13, total water use in Beijing shows an overall upward trend, accompanied by significant adjustments in the water use structure. Urban domestic water use and water use by the tertiary industry increase most significantly and gradually become the main sources of water resource consumption. Ecological water use maintains steady growth, whereas agricultural water use continues to decline. This change indicates that Beijing's water resource utilization pattern is shifting from a production-oriented mode to a domestic- and service-oriented mode.

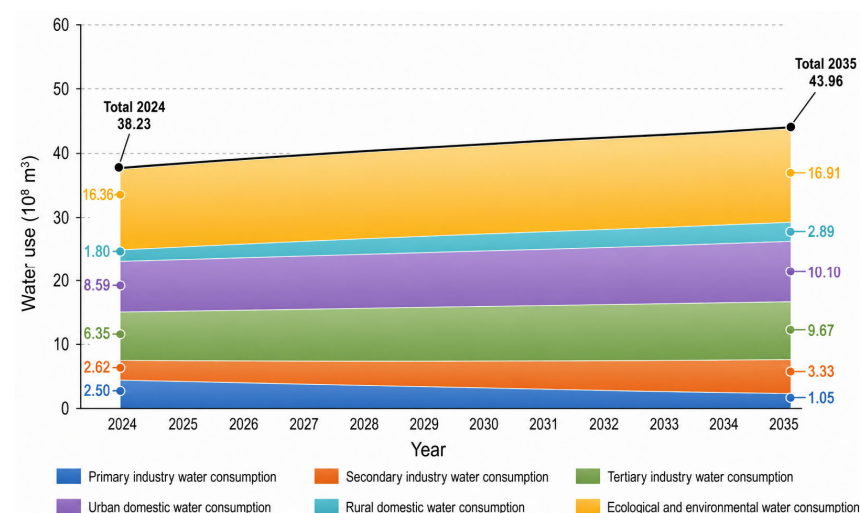


Figure 13. Simulated changes in water use in Beijing from 2024 to 2035. Note: The black line indicates the total water use.

4.2.3. Evolutionary Characteristics of Pollutant Discharge

As shown in Figures 14 and 15, COD and ammonia nitrogen emissions in Beijing exhibit consistent evolutionary trends during the study period, characterized by continu-

ous growth in total emissions and a significant transformation in the emission structure. In terms of total quantity, pollutant emissions show an overall upward trend, indicating that pollutant generation continues to accumulate under the combined effects of population growth and the expansion of economic activities. The reduction effect brought by improvements in wastewater treatment capacity is, to some extent, offset by the growth in demand.

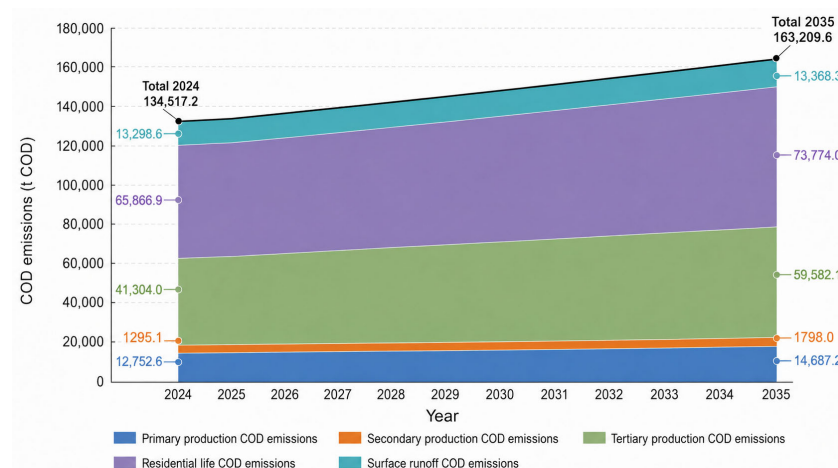


Figure 14. Simulated changes in COD emissions in Beijing from 2024 to 2035. Note: The black line indicates the total COD emissions.

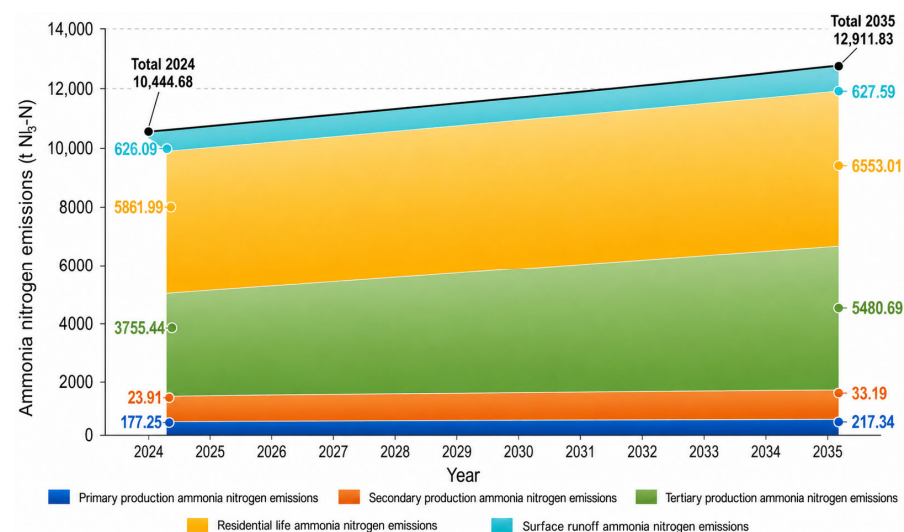


Figure 15. Simulated changes in ammonia nitrogen emissions in Beijing from 2024 to 2035. Note: The black line indicates the total ammonia nitrogen emissions.

According to the changes in emission structure generated by the model, the proportions of emissions from domestic sources and the tertiary industry continue to increase, while the proportion of industrial emissions declines relatively. This suggests that the dominant source of pollution is shifting from industry to domestic activities and the service sector. It also indicates that the development of the service sector amplifies water environmental pressure, even as the relative contribution of industrial pollution decreases.

From the perspective of evolutionary mechanisms, this transformation is mainly driven by urbanization and industrial restructuring. Pollutant discharge gradually shifts from concentrated industrial point sources to more dispersed domestic and service-sector sources, which significantly increases the complexity of pollution control. Therefore,

Beijing's water environmental pressure is shifting from "industrial pollution control" to "integrated urban pollution management".

4.3. Dynamic Assessment Results of the Urban Water Metabolism System

Based on the simulation results, the D-PI dynamic assessment method constructed in Section 3 was further applied to comprehensively analyze the coordination level and carrying status of Beijing's water metabolism system from a system-wide perspective. Specifically, this section presents the results of the coupling coordination degree between socioeconomic development and the water environment, the water environmental carrying status, and the coordinated and sustainable development status of the system identified using the two-dimensional D-PI matrix.

4.3.1. Analysis of the Coupling Coordination Degree Between Socioeconomic Development and the Water Environment

The simulated and predicted indicators of Beijing's water metabolism system were input into the coupling coordination degree model to calculate the coupling coordination degree D between socioeconomic development and the water environment from 2024 to 2035, as well as the integrated efficacy values of the two subsystems. The results are shown in Table 7 and Figure 16.

Table 7. Coupling coordination degree between socioeconomic development and the water environment in Beijing from 2024 to 2035 under system simulation.

Year	Integrated Efficacy Value of Socioeconomic Subsystem, $s(x)$	Integrated Efficacy Value of Water Environment Subsystem, $e(y)$	Coupling Degree, C	Coordination Index, T	Coupling Coordination Degree, D
2024	0.489	0.785	0.972	0.637	0.787
2025	0.513	0.772	0.979	0.643	0.793
2026	0.534	0.762	0.984	0.648	0.799
2027	0.555	0.752	0.989	0.654	0.804
2028	0.577	0.742	0.992	0.660	0.809
2029	0.606	0.733	0.995	0.669	0.816
2030	0.627	0.724	0.997	0.676	0.821
2031	0.649	0.715	0.999	0.682	0.825
2032	0.671	0.707	1.000	0.689	0.830
2033	0.701	0.699	1.000	0.700	0.837
2034	0.723	0.692	1.000	0.708	0.841
2035	0.746	0.685	0.999	0.715	0.845

As shown in Figure 16, based on the model simulation results, the coupling coordination degree between socioeconomic development and the water environment in Beijing shows a steady upward trend from 2024 to 2035, increasing from 0.787 in 2024 to 0.845 in 2035. The system gradually evolves from the stage of moderate coordinated development to the stage of well-coordinated development, indicating that the coordination between socioeconomic development and the water environment system continues to improve.

The evolutionary process of the coupling coordination degree can be divided into three stages. In the first stage, from 2024 to 2026, the coupling coordination degree D remains within the range of 0.7–0.8, indicating that the system is in the stage of moderate coordinated development. During this stage, the integrated efficacy value of the socioeconomic subsystem, $s(x)$, is lower than that of the water environment subsystem, $e(y)$, showing a socioeconomic-lagged type. This indicates that the water environment system still has a certain capacity to support socioeconomic development.

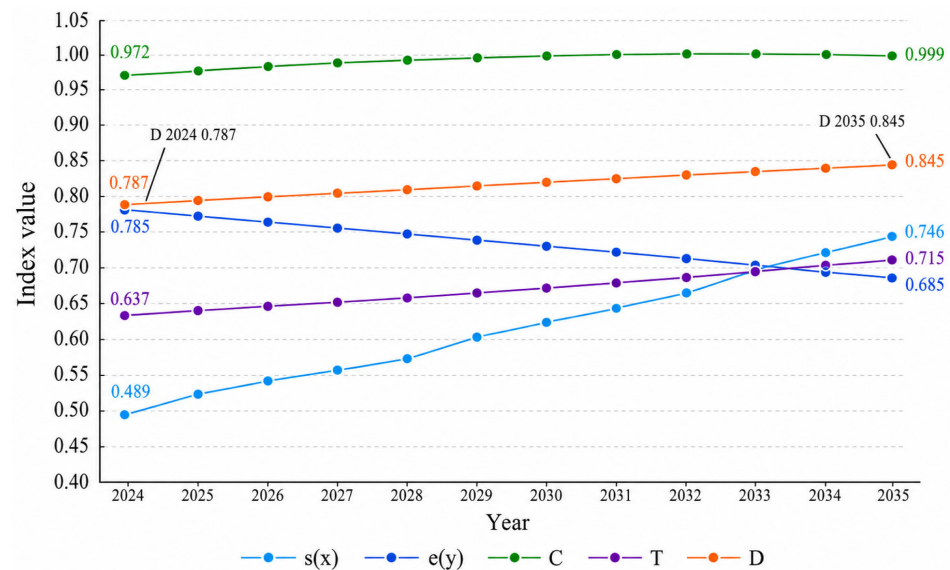


Figure 16. Variation trend of the coupling coordination degree between socioeconomic development and the water environment in Beijing from 2024 to 2035 under system simulation.

In the second stage, from 2027 to 2032, the coupling coordination degree first exceeds 0.8 in 2027, with $D = 0.804$, and the system enters the stage of well-coordinated development. During this period, although the overall coordination level of the system continues to improve, $s(x)$ remains lower than $e(y)$, indicating that socioeconomic development is still relatively lagging behind the water environment system. The improvement in system coordination is therefore mainly supported by the relatively favorable performance of the water environment subsystem.

In the third stage, from 2033 to 2035, the coupling coordination degree remains above 0.8, and the system stabilizes in the stage of well-coordinated development. Different from the previous stage, the integrated efficacy value of the socioeconomic subsystem, $s(x)$, exceeds that of the water environment subsystem, $e(y)$. The system therefore shifts from a “socioeconomic-lagged type” to a “water-environment-lagged type”, indicating that the water environment gradually becomes the key factor constraining further improvement in system coordination.

From the perspective of subsystem evolution, the integrated efficacy value of the socioeconomic subsystem, $s(x)$, shows a continuous upward trend, increasing from 0.489 to 0.746, with an average annual growth rate of approximately 2.3%. In contrast, the integrated efficacy value of the water environment subsystem, $e(y)$, declines slowly from 0.785 to 0.685. This result indicates that, during the study period, the pace of socioeconomic development continues to increase, leading to growing demand for water resource development and utilization as well as pollutant discharge. However, improvements in the water environment and governance capacity lag behind, resulting in a gradual increase in carrying pressure on the water environment subsystem.

Therefore, although the overall coordination of the system continues to improve, its internal structure has shifted from a “socioeconomic constraint” to a “water environmental constraint”. The water environment subsystem has gradually become the key bottleneck affecting further optimization of the urban water metabolism system. It should be noted that a higher D value does not necessarily mean lower environmental pressure. The D value reflects the relative coordination between the socioeconomic subsystem and the water environment subsystem, whereas the PI value directly reflects the absolute carrying pressure caused by water use and pollutant discharge. Therefore, the simultaneous

increase in D and PI indicates a non-synchronous pattern of coordination improvement and risk accumulation.

4.3.2. Calculation and Analysis of the Water Environmental Carrying Status

Based on the Nemerow index assessment method constructed in Section 3, the water resource carrying rate, COD carrying rate, ammonia nitrogen carrying rate, and comprehensive Nemerow index PI of Beijing from 2024 to 2035 were calculated. The results are shown in Table 8.

Table 8. Water environmental carrying status of Beijing from 2024 to 2035 under system simulation.

Year	Water Resource Carrying Rate	COD Carrying Rate	Ammonia Nitrogen Carrying Rate	PI
2024	1.238	1.941	2.545	2.249
2025	1.258	1.974	2.597	2.293
2026	1.269	2.018	2.657	2.343
2027	1.279	2.063	2.717	2.394
2028	1.289	2.107	2.776	2.443
2029	1.298	2.151	2.836	2.493
2030	1.305	2.194	2.894	2.542
2031	1.312	2.236	2.952	2.589
2032	1.319	2.277	3.008	2.636
2033	1.324	2.317	3.064	2.681
2034	1.329	2.356	3.117	2.725
2035	1.333	2.392	3.169	2.768

The results show that the water environmental carrying status of Beijing continues to worsen from 2024 to 2035. The comprehensive Nemerow index PI remains greater than 1.5 throughout the study period and increases from 2.249 in 2024 to 2.768 in 2035. This indicates that the water environment system remains in a state of severe overload over the long term, and that the degree of overload continues to deepen.

In terms of individual carrying indicators, the water resource carrying rate, COD carrying rate, and ammonia nitrogen carrying rate are all consistently greater than 1, indicating that the water environment system is under long-term pressure from both resource development and pollutant discharge. Among them, the ammonia nitrogen carrying rate remains the highest, increasing from 2.545 to 3.169, which indicates that ammonia nitrogen pollution is the key factor constraining the improvement of water environmental carrying capacity. This result may be related to the increasing contribution of domestic sewage and tertiary-industry wastewater under continued urbanization and service-sector expansion. In addition, ammonia nitrogen removal usually requires more stable biological treatment conditions than COD removal. Therefore, even with improvements in wastewater treatment capacity, ammonia nitrogen may remain the dominant carrying constraint. The COD carrying rate increases steadily from 1.941 to 2.392. By contrast, the water resource carrying rate shows a relatively small increase, but it remains in an overloaded state throughout the study period.

Overall, with population growth, the expansion of domestic water use, and the development of the tertiary industry, water demand and pollutant discharge continue to increase. However, improvements in water environmental capacity and governance capacity remain relatively limited, leading to the continuous accumulation of carrying pressure in the system. This indicates that Beijing's water metabolism system exhibits a prominent feature of "continuous accumulation of carrying pressure" during the study period and that water environmental constraints have become the core limiting factor affecting system sustainability.

4.3.3. Analysis of the Two-Dimensional D–PI Joint Identification Results

According to the dynamic changes in the coupling coordination degree D and the Nemerow index PI of Beijing from 2024 to 2035, the coordinated and sustainable development level of the urban water metabolism system can be comprehensively identified, as shown in Table 9.

Table 9. Identification results of the coordinated and sustainable development level of Beijing’s water metabolism system from 2024 to 2035.

Year	Coupling Coordination Degree, D	Nemerow Index, PI	Zone Classification
2024	0.787	2.249	■
2025	0.793	2.293	■
2026	0.799	2.343	■
2027	0.804	2.394	■
2028	0.809	2.443	■
2029	0.816	2.493	■
2030	0.821	2.542	■
2031	0.825	2.589	■
2032	0.830	2.636	■
2033	0.837	2.681	■
2034	0.841	2.725	■
2035	0.845	2.768	■

The results indicate that Beijing’s water metabolism system exhibits phased evolutionary characteristics during the study period. Before 2027, the system is generally located in the “Pressure–Adjustment Zone”. During this stage, the coupling coordination degree remains at the level of moderate coordinated development, indicating that a certain coordination capacity has been established between socioeconomic development and the water environment system. However, the Nemerow index remains within the severe overload range, suggesting that the water environment still faces substantial carrying pressure. The system as a whole is therefore exposed to a relatively high level of risk, and pollution control and resource regulation measures need to be strengthened urgently to alleviate environmental pressure.

From 2027 to 2035, as the coupling coordination degree continues to increase and enters the stage of well-coordinated development, the system gradually shifts into the “Warning–Optimization Zone”. This stage indicates that the coordination capacity between socioeconomic development and the water environment has improved significantly, but the carrying pressure on the water environment remains high and has not been fundamentally alleviated. Although the system has developed a relatively strong foundation for coordinated development, it is still significantly constrained by the water environment.

Overall, Beijing’s water metabolism system shows an evolutionary trend from “pressure–constraint dominance” toward “coordination–optimization dominance” during the study period. Nevertheless, water environmental carrying pressure persists, and the system has not yet entered a coordinated and sustainable development status. By combining ABM–SD simulation results with D–PI dynamic assessment, this study realizes an integrated analysis of “system evolutionary process simulation–operational state identification”, which represents an important extension of traditional static evaluation methods.

The non-synchronous evolutionary pattern of “coordination improvement–risk accumulation” can be explained by the different implications of D and PI . The coupling coordination degree D reflects the relative coordination between the socioeconomic subsystem and the water environment subsystem, which improves with industrial restructuring, water use efficiency improvement, and relatively high wastewater treatment capacity. In

contrast, PI reflects the absolute pressure of water resource use and pollutant discharge on the water environmental carrying capacity. During the simulation period, the continued growth of urban domestic water demand, ecological water use, and service-sector activities increases total water demand and pollutant loads. Therefore, the improvement in coordination does not necessarily indicate a reduction in carrying-capacity risk. This suggests that urban water management should combine coordination assessment with carrying-pressure monitoring.

5. Conclusions and Policy Recommendations

5.1. Conclusions

Based on the ABM–SD-coupled model and the D–PI dynamic assessment method, this study simulated and comprehensively evaluated Beijing’s urban water metabolism system. The results show that: (1) The evolution of the system is characterized by a non-synchronous process in which coordination improvement coexists with carrying pressure accumulation. During the study period, the coupling coordination degree gradually increased from the stage of moderate coordinated development to the stage of well-coordinated development, indicating that the coordination between socioeconomic development and the water environment system continued to improve. However, the Nemerow index remained within the severe overload range and showed an upward trend, suggesting that water environmental carrying pressure continued to accumulate. (2) The dominant constraint of the system gradually shifted from the socioeconomic subsystem to the water environment subsystem. Although the integrated efficacy value of the socioeconomic subsystem increased continuously, the integrated efficacy value of the water environment subsystem declined slowly, indicating that the water environment gradually became the key bottleneck restricting further optimization of the urban water metabolism system. (3) Pollutant discharge, especially ammonia nitrogen and COD emissions, was the core factor affecting the water environmental carrying status. The ammonia nitrogen carrying rate remained the highest among the carrying indicators, while the COD carrying rate also showed a continuous increasing trend. These results indicate that Beijing’s water metabolism system has not yet reached a coordinated and sustainable development status, despite the continuous improvement in system coordination.

5.2. Policy Recommendations

According to the above conclusions, the optimization and regulation of the urban water metabolism system should be promoted from the dual objectives of improving system coordination and alleviating carrying pressure. The following policy recommendations are proposed: (1) Pollutant source control and end-of-pipe treatment should be strengthened, with particular emphasis on targeted emission reductions in ammonia nitrogen and COD. As pollutant sources shift from industrial discharge to more dispersed domestic and service-sector sources, relying solely on end-of-pipe treatment is insufficient. Therefore, integrated measures, including source control, wastewater treatment improvement, ecological purification, and reclaimed water utilization, should be promoted. (2) The water use structure and resource allocation should be optimized. The utilization level of reclaimed water and overall water use efficiency should be improved, while industrial restructuring should be guided toward low-water-consumption and high-efficiency development to reduce the intensity of water resource development. (3) A differentiated regulation mechanism should be established for different agents. Residential water use should be mainly regulated through price constraints, whereas enterprise pollutant discharge should be controlled by strengthening cost constraints, thereby forming a coordinated water-saving–emission-reduction regulation pathway. (4) The coupling coordination degree (D) and the Nemerow

index (PI) should be incorporated into the water resource management evaluation system. A dynamic monitoring and early-warning mechanism with the dual objectives of coordination and carrying capacity optimization should be established to support continuous optimization of system operation. (5) Cross-departmental collaborative governance should be strengthened, and urban water management should shift from single-factor management toward integrated system governance, so as to promote long-term coordination among resource utilization, environmental protection, and economic development.

In line with the simulation results, nitrogen removal should be prioritized in wastewater treatment upgrading, because ammonia nitrogen shows the highest carrying rate. Refined management of urban domestic sewage and service-sector wastewater should also be strengthened, as domestic and tertiary-industry-related pressures continue to increase. In addition, reclaimed water reuse, demand-side water saving, leakage control, and water-use structure optimization should continue to be promoted because the water resource carrying rate remains above 1 throughout the simulation period.

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